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Sodium Alginate–Montmorillonite Composite Film Coatings for Strawberry Preservation

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Abstract: In this study, we prepared sodium alginate (SA) and montmorillonite (MMT) composite films for application in coatings for strawberry preservation. SA and MMT were used as the matrix and glycerol was used as a plasticizer. Six types of composite films with different MMT contents were compared by analyzing their mechanical properties, permeability, and preservation effects. The results show that the mechanical properties of the 10 and 20% MMT composite films were superior, with tensile strength and fracture elongation values reaching 63.09 and 48.06 MPa and 5.75 and 6.47%, respectively. Increased MMT content caused the water vapor permeability to decrease, while the effect on oil permeability was the opposite. A comparison of the preservation effect provided by the coatings showed that, on day 12, the weight loss, malondialdehyde content, and respiratory intensity of strawberries treated with the 20% MMT coating liquid decreased by 43.3, 25.8, and 57.1%, respectively, compared with the control. The contents of titratable acid, soluble sugar, total phenols, and soluble solids decreased by 25.8, 37.7, 25.9, and 14.5%, respectively. The results provide data support for the application of these new composite films as edible coatings for fruit preservation.

Keywords: sodium alginate; montmorillonite; composite film; strawberry preservation

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1. Introduction

Sodium alginate (SA) is a polysaccharide formed from β-D-mannuronic acid and α-L-guluronic acid via (1→4) glycoside linkages. SA is usually a white or light yellow powder that has been described as almost odorless and tasteless [1]. As an efficient coating preservative, SA is highly valued for its excellent stability and strong mechanical properties [2]. Additionally, it has excellent water solubility, film-forming ability, and safety, making it an ideal choice for food preservation. For example, SA has been applied in the preservation of various foods such as sweet cherries [3], strawberries [4], freshly cut apples [5], meat [6], and aquatic products [7]. SA effectively inhibits microorganism growth and delays food spoilage, making it a highly favored preservative for many applications [8,9].

Montmorillonite (MMT) is a type of non-metallic mineral material [10] with a fine scale-like appearance that is mostly white or grayish white. It is a layered silicate crystal composed of two layers of silica tetrahedron and aluminum oxide octahedron, which are prone to delamination. When the MMT interlayer is peeled off and increases in the dispersion system, this can improve the uniform dispersion of polymer-based composite materials, thereby significantly promoting the material's final properties [11,12]. In addition to its applications in materials science, MMT is also widely used in the food, medicine, and chemical industries, among others, due to its excellent water solubility, expansion capacity, barrier properties, non-toxicity, and ion exchange ability [13–16].

By creating a microenvironment with gas composition regulation and anti-corrosion functions, coating preservation technology provides an efficient means of food preservation. This easy-to-operate and cost-effective technology has been widely used in the field of fruit storage [17,18]. In this study, we examined the film-forming properties of SA and MMT. The effects of the coating on strawberries' weight loss rate, respiratory intensity, and malondialdehyde (MDA), titratable acid, soluble sugar, total phenol, and soluble solid contents were compared.

2. Materials and Methods

2.1. Preparation Process

2.1.1. Preparation of Composite Films

MMT with a mass ratio of above 95% was obtained from Zhejiang Hongyu New Materials Co., Ltd. (Huzhou, China) and purified following the protocol outlined in a previous study [19]. Diverse compositions of MMT and 50 mL of 2% (v/v) acetic acid solution were added to a three-necked flask and stirred evenly. This was followed by mixing in 2.0 g of SA (viscosity: 200 ± 20 mpa.s) (Shanghai Aladdin Biochemical Technology Co., Ltd., Shanghai, China) and 100 mL of 2% (v/v) acetic acid solution. Next, 10% glycerol (calculated using SA mass) was added and the mixture was uniformly stirred at 60 °C in the three-necked flask. A composite film solution was prepared using ultrasound after degasification. We used this solution to coat a polyethylene terephthalate plate and form a film, which was then dried at 50 °C in an oven. Finally, the film was removed and stored in a dryer for later use. The film's components, such as MMT, SA, and glycerol, are all eco-friendly and meet food-quality and contact-material requirements.

2.1.2. Strawberry Preservation

Fresh strawberries without mechanical damage were selected for the experiments, with 36 strawberries in each of the four groups. The control group and three treatment groups were as follows: fresh strawberries without coating (CK); fresh strawberries coated with SA film solution (T1); fresh strawberries coated with 10% MMT composite film solution (T2); and fresh strawberries coated with 20% MMT composite film solution (T3). After coating the treatment group strawberries [13], the fruit was hung with a clamp and air-dried. After drying, the strawberries in each group were stored at 4 °C for 12 days. The strawberries' mass loss rate, respiratory intensity, and MDA, titratable acid, soluble sugar, total phenol, and soluble solid content were measured and compared at days 0, 3, 6, 9, and 12 of storage.

2.2. Testing Method

2.2.1. Mechanical Properties

The thickness of each composite film was determined by averaging measurements from five different spots using a micrometer caliper (±0.01 mm). The films were then cut into 80 × 20 mm^2 rectangles for tensile testing. Their tensile strength (T_S) and elongation at break (E) were measured using a physical property tester (TA.XT Plus, Stable Micro Systems, London, UK). For each group, we measured three transverse and three longitudinal samples in the vertical direction, obtaining a total of six parallel samples, and the average T_S and E were calculated. The spacing was 80 mm and the sample speed was 100 mm/min. To calculate T_S, the following formula was used:

$$T_S = F/S \quad (1)$$

where T_S is the tensile strength (MPa), F is the tensile force of the film (N), and S is the film's cross-sectional area (m^2). The formula to calculate E is as follows:

$$E = \left(\frac{L}{L_0} - L_0\right) \times 100\% \quad (2)$$

where E is the fracture elongation (%), L_0 is the original standard distance of the sample (mm), and L is the standard distance of the sample fracture (mm).

2.2.2. Permeability

Water vapor permeability (WVP): The WVP of the composite films was measured using the quasi-cup method [20,21]. Briefly, we coated a cell containing a certain amount of dry $CaCl_2$ with the composite film and sealed it with melted wax. Then, the cell was placed in a 100% humidity environment at 25 °C. Next, the cell weight was measured every 24 h for 1 week. The WVP was defined as the amount of water vapor transported through the film per unit of time, pressure, and film area; it was calculated using the following equation:

$$WVP = \Delta m \times d/(A \times t \times \Delta P) \quad (3)$$

where WVP is the water permeability coefficient ($g \cdot mm/m^2 \cdot d \cdot KPa$), Δm is the increase in cell weight (g), d is the thickness of the film (mm), A is the effective area of the film (m^2), t is the time interval of the measurement (d), and ΔP is the vapor pressure difference on both sides of the sample (KPa).

Oil permeability coefficient (OPC): The oil permeability of the composite films was measured following the protocol outlined in a previous study [20] with slight modification. First, a tube containing 5 mL of peanut oil with samples was sealed. Then, the tube was inverted onto filter paper. Next, the weight of the filter paper was measured for 1 week to check the OPC of the samples. The OPC was calculated using the following equation:

$$OPC = \Delta m \times d/A \times T \quad (4)$$

where OPC is the oil permeability coefficient ($g \cdot m/m^2 \cdot d$), Δm is the weight increase in the filter paper (g), d is the thickness of the composite film (mm), A is the effective area of the composite film (m^2), and T is the time taken for oil permeation (days).

2.2.3. Fourier Transform Infrared Spectroscopy (FTIR)

To conduct FTIR, 1 mg of a composite film sample combined with 100 mg of KBr was ground in a mortar. Then, using a mold, the ground mixture was compressed into a disk shape with a diameter of 13 mm and a thickness of 1 mm. Using an FTIR spectrometer (FTIR-650, Bruker, Karlsruhe, Germany) to scan the samples from 4000 to 500 cm^{-1}, we produced a spectrum at a resolution of 4.0 cm^{-1} over 64 scans.

2.2.4. Weight Loss

The strawberries of each group were individually weighed every 3 days [22], and the weights were recorded. The strawberry mass loss rate was measured in % and calculated according to the following formula:

$$\text{Weight loss } (\%) = \frac{W_0 - W_t}{W_0} \times 100 \quad (5)$$

where W_0 and W_t are the initial weight and the weight at time t, respectively.

2.2.5. MDA Content

The MDA content of the strawberries in each group was determined based on the method outlined by Zhang et al. [23], with slight modification. First, 10 g of strawberries were crushed using a grinder. Then, 20 mL of 10% (*w*/*v*) trichloroacetic acid (TCA) was added to 2 g of each sample group, and the mixture was ground in an ice bath. The mixtures were then centrifuged at 4000 r/min for 10 min at 4 °C, and 1 mL of supernatant was transferred to 1 mL of 10% (*w*/*v*) TCA containing 0.67% (*w*/*v*) thiobarbituric acid; then, this was incubated in boiling water for 15 min. After rapid cooling, the mixture was centrifuged at 4000 r/min for 10 min. Supernatant absorbance was measured at 532 nm,

600 nm, and 450 nm using a spectrophotometer. The control was 10% TCA. The MDA content was reported as μmol/g of fresh weight and calculated using the following formula:

$$\mathrm{MDA}(\mu\mathrm{mol/g}) = 6.45 \times (A_{532} - A_{600}) - 0.56 \times A_{450} \quad (6)$$

where A_{532}, A_{600}, and A_{450} are the absorbance of the supernatant at wavelengths of 532 nm, 600 nm, and 450 nm, respectively.

2.2.6. Respiratory Intensity

A 0.4 mol/L NaOH solution (20 mL) was added to a culture dish [24] and placed in the center of a dryer. Then, 100 g of strawberries were weighed, sealed, and placed in a dryer. After 1 h, the alkaline solution was transferred from the culture dish into an iodine volumetric flask and rinsed four to five times with distilled water. Then, 5 mL of saturated $BaCl_2$ solution and two drops of 1% phenolphthalein were placed into the flask. In order to produce a blank control, the mixture was titrated with 0.2 mol/L oxalic acid solution until a milky white color appeared and did not fade for 30 s. The respiratory intensity was measured in mg/(kg·h) using the following equation:

$$\text{Respiratory intensity} = \frac{(V_1 - V_2) \times M \times 44}{W \times H} \quad (7)$$

where M is the molar concentration of oxalic acid, mol/L; V_1 is the amount of blank titration, mL; V_2 is the sample drop quantification, mL; W is the sample weight, kg; H is the determination time, h; and 44 is the molecular weight of carbon dioxide.

2.2.7. Titratable Acid Content

According to the GB/T12456-2008 specifications [25], strawberries from each sample group were mashed, and a certain amount of the slurry was weighed and placed into a 100 mL beaker. The contents of the beaker were then transferred into a 250 mL volumetric bottle with distilled water at about 80 °C. The total volume of the mixture was about 150 mL. Then, the mixture was boiled in a water bath for 30 min, shaken 2 to 3 times during the boiling period, and then cooled to room temperature, with the final volume adjusted to 250 mL. Next, the mixture was centrifuged at a speed of 4000 r/min for 10 min to obtain a filtrate. A 25 mL volume of the filtrate was titrated in a small beaker with 0.1 mol/L NaOH solution until the pH reached 8.20, and the volume of NaOH solution consumed was recorded as V_1. Meanwhile, the volume of NaOH solution consumed was recorded as V_2. The titratable acid content was measured in % and calculated using the following formula:

$$\text{Titrable acid content} = \frac{V \times C \times (V_1 - V_2) \times 0.064}{V_s \times m} \times 100 \quad (8)$$

where V is the total volume of the sample extract, mL; C is the molar concentration of NaOH solution, mol/L; 0.064 is a constant applicable for 1 mL of 0.1 mol/L NaOH solution and equivalent to the number of grams of citric acid, g/mL; V_S is the sample size taken during titration, mL; and m is the mass of the sample, g.

2.2.8. Soluble Sugar Content

Employing the anthrone method [26], 100 μg/mL of standard sucrose solution, at concentrations of 0, 0.2, 0.4, 0.6, 0.8, and 1.0 mL, was utilized to obtain sucrose solutions with concentrations of 0, 20, 40, 60, 80, and 100 μg/L. The solutions were poured into test tubes, and distilled water was used as a blank control. Then, 5 mL of anthrone–sulfuric acid solution was poured into each test tube. After full oscillation, the test tubes were placed into a boiling water bath, with each tube held for 1 min, and then allowed to cool to room temperature after removal. The absorbance value of the reaction liquid was measured at 620 nm, and the standard curve was drawn with the absorbance value as the vertical coordinate and the sucrose mass as the horizontal coordinate.

A 0.5 mL volume of sample extract was transferred into a 25 mL scale test tube; then, 1.5 mL of distilled water was added and mixed evenly. Next, 5 mL of anthrone–sulfuric acid solution was added, and the test tube was fully shaken and then placed into a boiling water bath, with the tube held for 1 min, and then allowed to cool to room temperature after removal. The absorbance of the reaction solution was quantified at 620 nm, and the soluble sugar content was quantified in mg/g using the following equation:

$$\text{Soluble sugar content} = \frac{\mu \times V}{W} \quad (9)$$

where μ is the concentration of soluble sugar obtained from the standard curve, μg/mL; V is the constant volume of the sample, mL; and W is its weight, g.

2.2.9. Total Phenol Content

The total phenol (TP) content was measured with gallic acid as the standard sample according to a modified Folin–Ciocalteu method [27]. The crushed samples (2 g) and 20 mL of 70% (*v*/*v*) ethanol were combined, and ultrasonic extraction was performed for 90 min at 50 °C. Then, the mixture was centrifuged at 4000 r/min for 10 min at 4 °C. Next, the supernatant was properly diluted and then mixed with 1 mL of Folin–Ciocalteu reagent. After 2 min of incubation at room temperature, 4 mL of Na_2CO_3 (10.6%, *w*/*v*) was added, and the resulting mixture was incubated for 30 min at room temperature. At the end of the incubation period, the absorbance was measured at 760 nm. The total phenol content was expressed as the mass of gallic acid equivalent in fresh strawberry weight (mg/g).

2.2.10. Soluble Solid Content

To determine the soluble solid content for each sample group, the distilled water was adjusted to zero; then, a certain amount of strawberry sample was weighed using an electronic balance [28]. Next, each sample was ground into a homogenate in a mortar. One drop of strawberry slurry was dropped into the measuring chamber of a digital display saccharometer to determine the content of soluble solids in each group. Each group was tested three times, the display data were recorded, and the average value was determined.

2.2.11. Data Processing

Excel 2019 was used for statistical analysis and Origin 2019 software facilitated the creation of maps. SPSS 22.0 software was employed to assess the significant differences and $p < 0.05$ indicated a statistically significant difference.

3. Results

3.1. Film Properties

3.1.1. Mechanical Properties

Figure 1 shows the mechanical properties of composite films with varying MMT contents. T_S and E at break are important indexes for evaluating the ductility and flexibility of thin film materials (with thicknesses of about 0.1 mm). The experimental results showed that the T_S of the composite films initially increased and then decreased with increased MMT content. When the MMT content was 12%, the T_S was optimal, reaching 67.77 MPa. The reason for this was that the large specific surface area of the MMT layer allowed the formation of hydrogen bonds and secondary bonds between MMT and SA molecules, promoting their uniform and stable dispersion in the matrix during the filling process. When composite film is subjected to external forces, the stress can be effectively transferred from the SA matrix to the MMT matrix, thus enhancing the overall T_S [29]. However, as the MMT content continued to increase, the T_S of the composite film decreased. While an appropriate amount of MMT can strengthen the mechanical properties of the composite film, excessive MMT incorporation can reduce the film's plasticity, ultimately compromising mechanical strength.

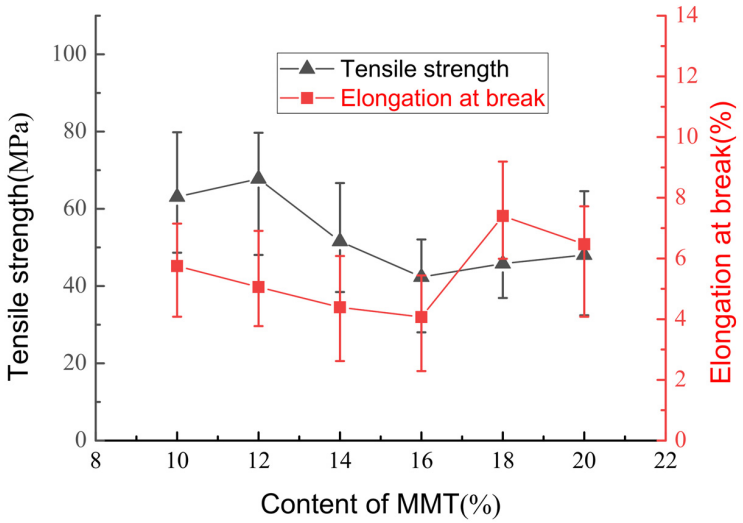


Figure 1. Effect of montmorillonite (MMT) content on the mechanical properties of the composite films.

With increased MMT content, the elongation at break initially decreased, subsequently rose, and then declined. The lowest point was 4.07% at 16% MMT content. The hydrogen bond interactions between MMT and SA molecules created a strong association, which, combined with MMT's multi-layer structure and large specific surface area, restricted the freedom of motion among the SA molecular chains. The brittleness of the composite film increased, resulting in a reduced E [30,31]. When the MMT content reached 20%, its filling effect in the composite film became more pronounced, and the originally increased elongation at break decreased again. Therefore, in the process of preparing the composite film, it was necessary to precisely control the additional proportion of MMT in order to increase the E as much as possible while maintaining the T_S. In this way, we could optimize the overall mechanical properties of the film samples [32].

The mechanical properties of the 10% and 20% MMT composite films were determined to be better than the other films, and the T_S and elongation at break reached 63.09 and 48.06 MPa, and 5.75 and 6.47%, respectively.

3.1.2. Permeabilities

Figure 2 illustrates the influence of MMT content on the water vapor and oil permeabilities of the film samples. WVP is a measure of the water exchange efficiency between membrane materials and the external environment. A lower WVP value indicates a composite film with better water vapor barrier performance. The WVP of the tested film samples with different MMT contents fluctuated from 10% (2.90 $g \cdot (m^2 \cdot h)^{-1}$) to the lowest point at 20% (1.81 $g \cdot (m^2 \cdot h)^{-1}$). The intercalation reaction that occurred after MMT and SA were mixed resulted in the participation of many surface hydroxyl groups, forming a surface primarily composed of oleophilic organic groups. At the same time, there were orderly dispersed nanoparticle layers in the polymer matrix. On the one hand, with increased MMT content, the degree of reaction with SA was increased and SA hydrophilicity was reduced via binding with the surface hydroxyl group [33,34]. On the other hand, due to the formation of the nanoparticle layer, water vapor had to traverse a zigzag path around the polymer matrix, effectively increasing its diffusion path length across the film and limiting swelling capacity. Consequently, the diffusion of water molecules in the membrane was limited, which enhanced the film's water vapor barrier performance [35].

The OPC of the film samples exhibited an increase with rising MMT content. It was determined that 10% MMT can achieve effective dispersion in the SA substrate. The OPC was 0.0249 $g \cdot m/(m^2 \cdot h)$, forming a uniform and dense polymer film with SA, which helped to improve the barrier property of the composite film. However, with increased MMT content, more surface hydroxyl groups were consumed in the intercalation reaction

with SA [33]. Additionally, the formation of excessive agglomerations could compromise the microstructural integrity of the composite film, making it easier for oil molecules to penetrate the matrix. This may have negatively impacted the oil-blocking performance of the composite film [36], which reached a maximum OPC of 0.0648 g·m/(m^2·h).

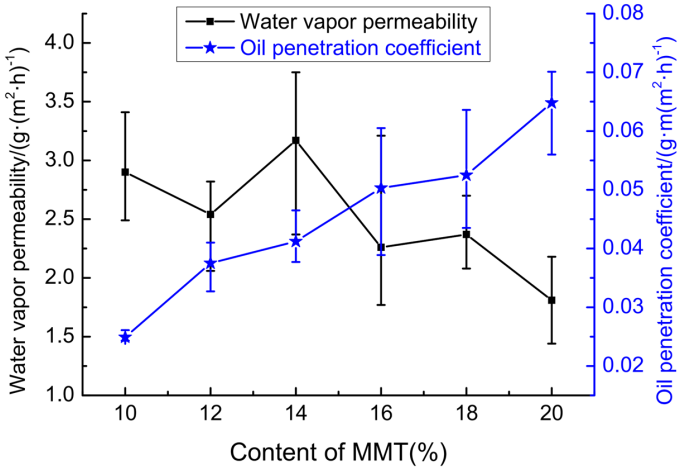


Figure 2. Effect of MMT content on the permeability of composite films.

3.1.3. FTIR

Figure 3 presents the infrared spectra of the film samples with varying MMT contents and the main characteristic absorption peaks. As illustrated in the figure, an absorption peak appeared at 3475 cm^{-1} for all composite films, which corresponded to the O-H stretching vibration of SA [37]. The positions of the absorption peaks presented by the seven composite films with different MMT contents were quite close, showing no significant difference in peak size or position between those with MMT (10%, 12%, 14%, 16%, 18%, and 20%) and those without (0%). This indicates that the addition of MMT did not introduce new chemical bonds [38]. The absorption peak of the composite film near 1610 cm^{-1} was attributed to the asymmetric stretching vibration of −COO−, accompanied by a redshift due to hydrogen bonding. The absorption peak near 1411 cm^{-1} corresponded to the symmetric stretching vibration of −COO−. The above phenomena indicate that only hydrogen bonds exist between the substances in the composite film, and no new substances were generated. Thus, MMT was uniformly dispersed in SA, forming the composite film.

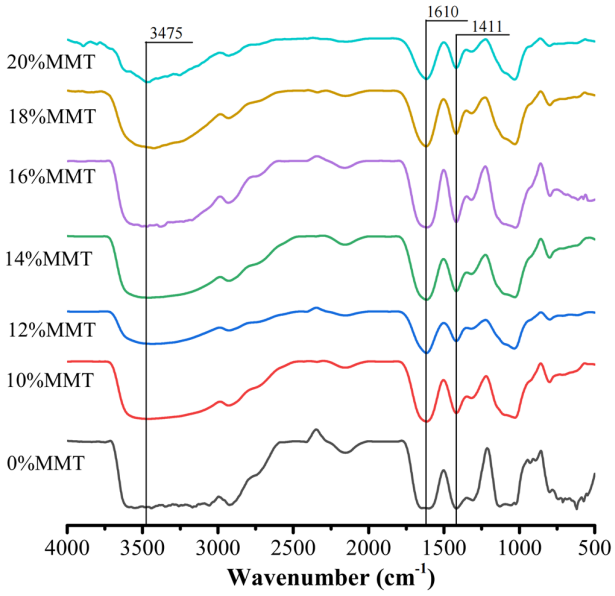


Figure 3. Infrared spectra of composite films with different MMT contents.

3.2. Strawberry Preservation Analysis

3.2.1. Weight Loss

Figure 4 shows the change in the weight loss rate of strawberry samples, in the three fresh treatment groups and the blank control group, over the storage duration. With the extension of the storage duration, the weight loss rate in all groups showed an increasing trend. After 12 days of storage, the weight loss values of the CK and T1 groups were 16.92% and 15.04%, respectively, indicating that the SA coating alone could not effectively reduce water loss. In the composite coating group, the weight loss (13.90% and 9.59%) significantly decreased ($p < 0.05$) with increased MMT content. The SA/MMT composite coating can effectively control strawberry metabolism and water loss, and the addition of MMT can significantly enhance the coating's water vapor barrier properties, thereby inhibiting fruit water loss and transpiration. Therefore, the addition of an appropriate amount of MMT can effectively reduce the weight loss in the fruit. Among the tested groups, the T3 group exhibited a 43.3% greater reduction in weight loss during storage compared to the CK group. Therefore, the T3 group significantly inhibited water loss in strawberries, demonstrating the coating's effectiveness as a fruit preservative [39,40].

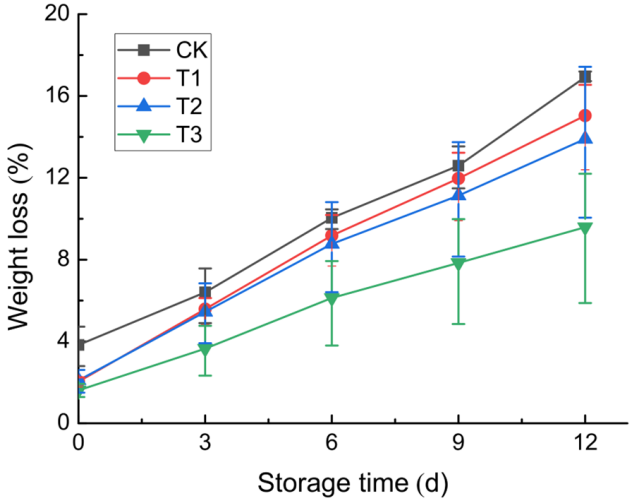


Figure 4. Effect of different composite film liquid coatings on strawberry weight loss.

3.2.2. MDA Content

A significant positive correlation was found between the MDA content and the lipid oxidation level in strawberries; the former can thus be identified as a key indicator in evaluating the membrane lipid oxidation level of strawberries during storage, which can also be used to evaluate epidermal cell integrity and fruit freshness [41].

Figure 5 shows the changes in the MDA content of strawberry samples over the storage duration for three preservation treatment groups and a blank control group. With the change in storage duration, the MDA content of strawberry fruit in each group showed an upward trend. After the coating treatment, the MDA content of strawberries (day 12: 23.10, 22.54, and 21.11 μmol/gFW) was lower in the three treatment groups than that of the CK group (28.45 μmol/gFW) ($p < 0.05$). Among them, the MDA content of the T3 group decreased by 25.8% when compared with the CK group, indicating that the protective film formed on the fruit's surface by the preservative coating solution effectively delayed ripening and aging. The addition of MMT significantly improved the oxygen barrier performance of the composite coating, effectively reducing the oxygen concentration around the strawberry fruit and slowing down the peroxidation rate of membrane lipids in epidermal cells [42].

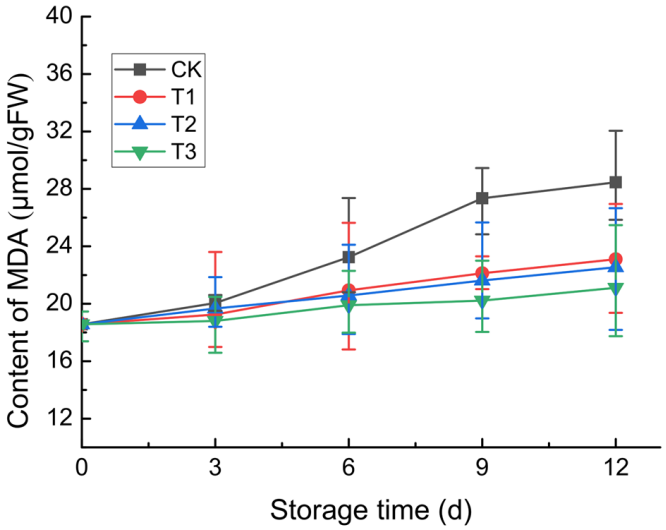


Figure 5. Effect of different composite film liquid coatings on the malondialdehyde content of strawberries.

3.2.3. Respiratory Intensity

Respiration is a key physiological process during postharvest fruit and vegetable storage. Its intensity directly reflects the metabolic rate and has an important impact on the quality and storage capacity of fruits and vegetables [43]. Research has shown that strawberries experience a significant decrease in their respiratory intensity before they ripen. However, in the post-ripening stage after picking, the respiratory intensity begins to gradually increase, especially in the first 2 to 3 days, before reaching the highest respiratory peak. Subsequently, the respiratory intensity gradually stabilizes and eventually decreases [44].

Figure 6 shows the changes in respiratory intensity of strawberry samples over the storage duration in the three fresh treatment groups and a control group. Overall, it showed a trend of first decreasing, then increasing, and finally decreasing. During the first 3 days, the respiratory intensity of all experimental groups showed a downward trend. This decline can be attributed to insufficient pre-cooling time when the strawberries were transferred from room temperature to storage temperature (4 °C), sharply slowing the metabolic process and making temperature the dominant factor affecting respiratory intensity. After reaching equilibrium on day 3, oxygen gradually diffused through the coating layer, increasing the available oxygen content of the fruit, which subsequently led to a gradual increase in metabolic activity and respiratory intensity. On day 12, the fruit respiratory intensities in the T1, T2, and T3 groups were significantly lower than in the CK group (88.96 mg/(kg·h)) ($p < 0.05$), measuring 44.97, 54.12, and 38.14 mg/(kg·h), respectively, and the T3 group's respiratory intensity was 57.1% lower than that in the CK group. With the addition of MMT, the permeation path of oxygen, carbon dioxide, and other gases was reduced, which enhanced the oxygen barrier performance of the composite film, resulting in a decrease in the oxygen level around the fruit. As the coating effectively closes the stomata, the availability of oxygen in fruit tissue respiration is also significantly reduced and effectively inhibited [45,46].

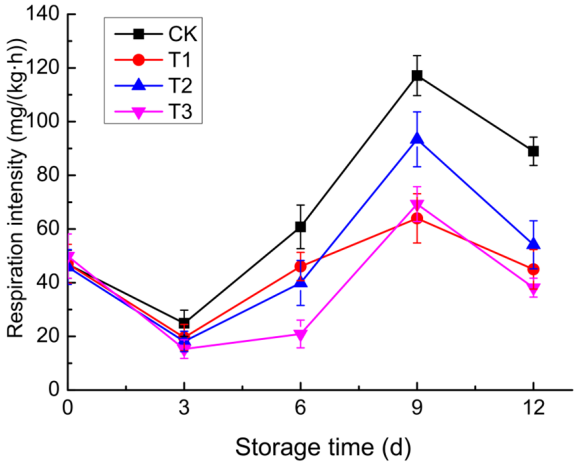


Figure 6. Effect of different composite film liquid coatings on the respiratory intensity of strawberries.

3.2.4. Titratable Acid Content

In fruits and vegetables, acid is not only a raw material in the respiratory metabolic process but also a key element in the development of their unique taste. Therefore, during storage, the change in the titratable acid content of strawberries can directly indicate the degree of nutrient consumption. At the early stage of storage, strawberries are not fully ripe and their respiratory intensity is weak. As respiratory raw materials, titratable acids begin to gradually accumulate and increase. However, at the later stage of storage, some of the titratable acid is consumed as a respiratory raw material, while the rest is converted into sugar, resulting in a downward trend in the overall content [47].

Figure 7 shows the change in titratable acid content over the storage duration for strawberry samples from the three fresh treatment groups and the control group. Among the groups, the titratable acid content of the CK group decreased most rapidly and was significantly lower than that of the experimental group treated with the composite coating at the later stage of storage (9–12 days) ($p < 0.05$). The titratable acid content of the CK group was 0.4787% on day 12 and about 71.5% lower than during early stage of storage, which was 25.8% higher than that of the T3 group. In the coating group, the titratable acid consumption in the T3 group was significantly lower than that in the T1 group ($p < 0.05$). At the end of the storage period, only 0.0241% of the titratable acid was consumed in the T3 group. These experimental results paralleled those for respiratory intensity, indicating that the effect was significant. Therefore, the coating could reduce titratable acid loss by slowing down respiration and delaying fruit ripening [48,49].

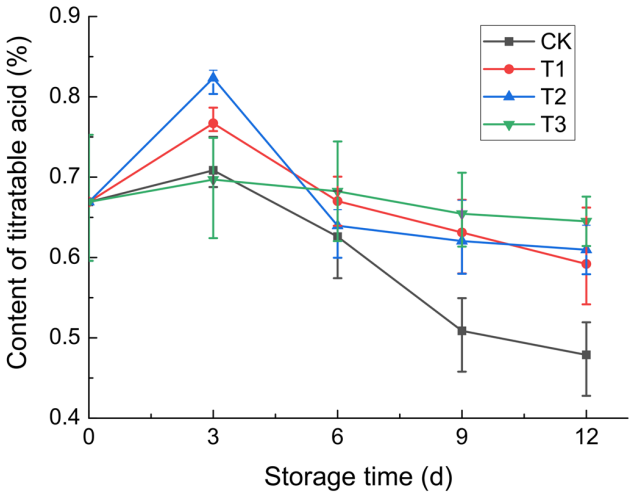


Figure 7. Effect of different composite film liquid coatings on titratable acid content of strawberries.

3.2.5. Soluble Sugar Content

Figure 8 shows the changes in soluble sugar content over the storage duration in strawberry samples from the three fresh treatment groups and the control group. Due to the large decrease in water during storage, the soluble sugar content generally decreased sharply at first and then more gradually. Soluble sugar is not only a key nutrient element in strawberry fruit but also an important respiratory substance. Therefore, changes in soluble sugar content can directly reflect the respiratory and metabolic conditions of strawberry fruit during storage [47]. The experimental results were generally consistent with the changes in respiratory intensity, and the changes in the treatment group were greater than those in the CK group. The soluble sugar content (18.02 mg/g) of the CK group on day 12 was about one-third of that at the beginning of storage. The soluble sugar content of the T3 group was the highest among the treatment groups, at 37.7% higher than that of the CK group. This result indicated that the protective film formed by the preservative coating liquid can effectively slow down the respiration rate, thus delaying the soluble sugar reduction rate [50]. Meanwhile, with increased MMT, the oxygen barrier property of the coating was further improved, the oxygen content around the fruit was reduced, and the coating closed the stomata. In turn, this reduced the respiratory rate, thus effectively inhibiting the respiratory action of strawberries and reducing soluble sugar consumption.

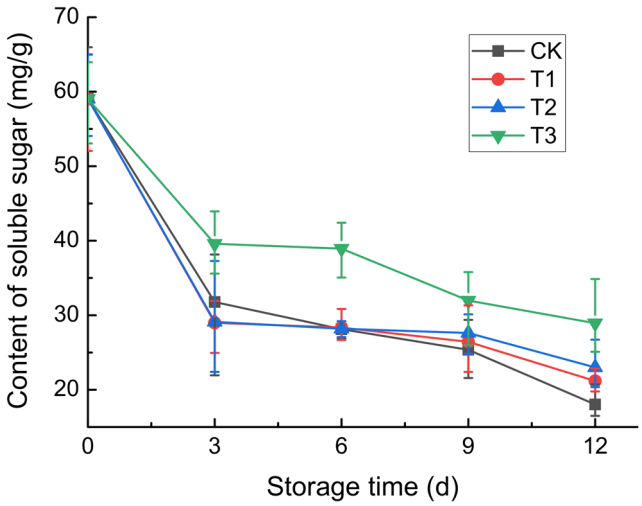


Figure 8. Effect of different composite film liquid coatings on the soluble sugar content of strawberries.

3.2.6. Total Phenol Content

Figure 9 shows the changes in the total phenol content over the storage duration for strawberry samples from the three fresh treatment groups and the control group. The total phenol content of the coating group first increased and then decreased, and that of the CK group showed a decreasing trend. The total phenol content of the samples after 0–3 days of treatment showed an increasing trend. Under the adverse conditions of low O_2 and high CO_2, after coating, reactive oxygen species were easily generated and accumulated in strawberries. In order to adapt to the coating environment and eliminate reactive oxygen species, antioxidant-reactive enzyme activities were increased. However, at the later stage of storage, as the strawberries began to age, the activity of antioxidant-active enzymes declined, excessive reactive oxygen species could not be effectively removed, and the total phenol oxidation rate accelerated, resulting in a gradual decrease in its content [47]. On day 12, the total phenol content of group CK (5.99 mg/g) was lower than that of the experimental group ($p < 0.05$), decreasing to 22.6% of the initial value (day 0), which was 25.9% higher than that of group T3. The total phenol content of the T3 group was the highest. Because the SA/MMT composite film has excellent oxygen-blocking properties, its low oxygen permeability significantly slowed down the respiration rate of postharvest

strawberries. Meanwhile, this composite film showed a positive regulatory effect on reactive oxygen metabolism, thereby reducing total phenolic antioxidant loss [51,52].

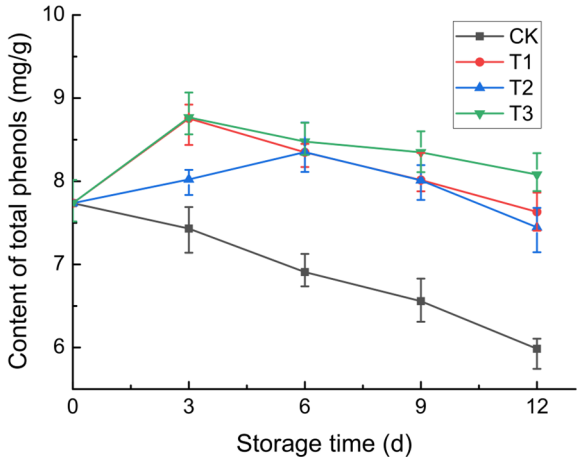


Figure 9. Effect of different composite film liquid coatings on the total phenol content of strawberries.

3.2.7. Soluble Solid Content

Soluble solids are mixtures of compounds such as sugars, acids, vitamin C, and certain water-soluble pectins. They can be used as a measure of the total sugar content of fruit and can also indicate ripeness. Figure 10 shows the changes in soluble solid contents over the storage duration for strawberry samples in the three fresh treatment groups and the control group. Throughout the storage period, the soluble solid content showed a trend of first increasing and then decreasing. At the beginning of storage, the strawberries had not yet reached the full maturity stage, so their respiratory intensity and nutrient consumption were relatively low. On day 12, the soluble solid content (10.0%) of the CK group was lower than that of the coated experimental groups ($p < 0.05$), and the soluble solid content (11.1%) of the T1 group was lower than that of the T2 (11.3%) and T3 groups (11.7%), because the coating treatment effectively reduced the metabolic rate. Subsequently, as the soluble solid content was kept stable, the ripening process was delayed. In contrast, the fruit metabolism of the control group was more vigorous, resulting in a greater consumption of soluble sugars and organic acids [53].

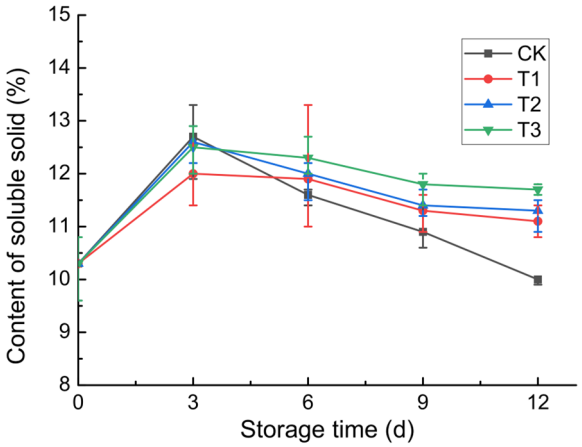


Figure 10. Effect of different composite film liquid coatings on the soluble solid content of strawberries.

Compared with other alginate-based coating solutions for strawberries, the shelf life of strawberries using such solutions is shorter than those coated with functional ingredients such as green tea extract [54], carvacrol and methyl cinnamate [40], and essential oils [55]. These functional ingredients provide antibacterial, antioxidant, and other beneficial effects.

However, the coating solution with an inorganic salts/alginate blend is significantly inferior to the MMT-based coating according to the data of respiration rate and total soluble solids [56,57]. This is likely because the layered structure of MMT complements the advantages of sodium alginate in terms of permeability and selectivity.

4. Conclusions

In this study, composite films were produced using SA and MMT as the primary materials. The mechanical properties, water vapor and oil permeabilities, and infrared spectra of the SA/MMT films were investigated, and their preservation effects on strawberries were evaluated. The mechanical properties of the 10% and 20% MMT composite films were found to be optimal, with T_S and E reaching 63.09 and 48.06 MPa, and 5.75 and 6.47%, respectively. The lowest WVP was 1.81 g/(m^2·h) at 20% MMT, and the lowest oil permeability was 0.0249 g·m/(m^2·h) at 10% MMT.

Our analysis of the weight loss, MDA content, respiratory intensity, titratable acids, soluble sugars, total phenols, and soluble solid content of strawberries coated with four different film solutions therefore demonstrated that the 10% and 20% MMT composite films had remarkable effects in maintaining freshness when compared with the untreated control. These results provide data supporting the application of the new composite films as edible coatings and fruit preservation materials.

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